

1. Consider a pendulum made of a spring with mass m on the end. The spring is arranged to lie on a straight line (which we arrange by, say, wrapping the spring around a rigid massless rod). The equilibrium length of the spring is l . Let the spring have length $l + x(t)$ and let the angle with the vertical be $\Theta(t)$. Assume the motion takes place in a vertical plane.



- (a) Write down the Lagrangian.
- (b) Write down the Euler-Lagrange equations.
- (c) Find the equations of motion for x and Θ .

(6 marks)

2. Given a point mass m with Lagrangian of the form:

$$L(r, \theta, \dot{r}, \dot{\theta}) = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{k}{r}$$

Find the Hamiltonian using Legendre Transforms.

(5 marks)

3. Show that

$$\frac{\partial^2 H}{\partial p^2} \cdot \frac{\partial^2 L}{\partial u^2} = 1$$

(3 marks)

4. A particle moves in a potential

$$V(r) = -V_0 e^{-(\lambda^2 r^2)}$$

- (a) Write down the effective potential for the system, for a given angular momentum ℓ .
- (b) Given ℓ , find the radius of the stable circular orbit. An implicit equation is fine.
- (c) It turns out that if ℓ is too large, then no circular orbit exists. What is the largest value of ℓ for which a circular orbit does in fact exist?

(6 marks)